ASU Research Computing
Arizona State University

# Tuning Workflow Management Systems for shared HPC resources

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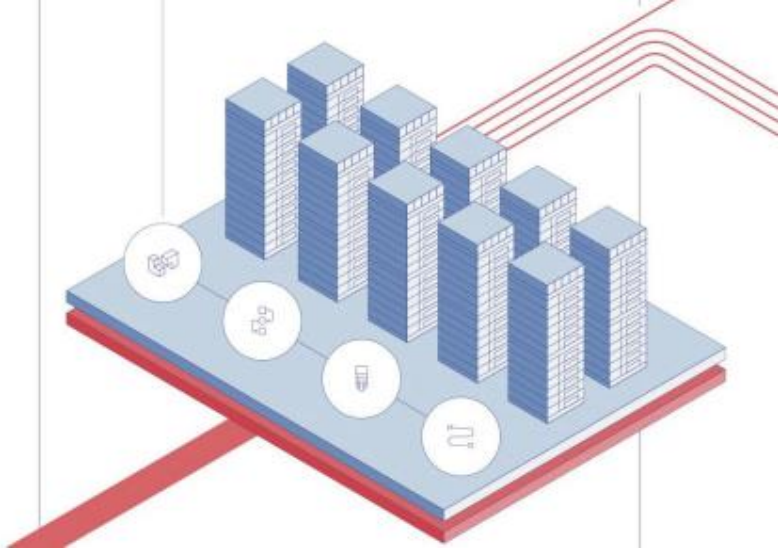
# Content

- Overview
- Popular Options
- Basic Pipeline Structure
- Brief Coding Examples
- Challenges
- Best Practices
- Benchmarking

# Keywords

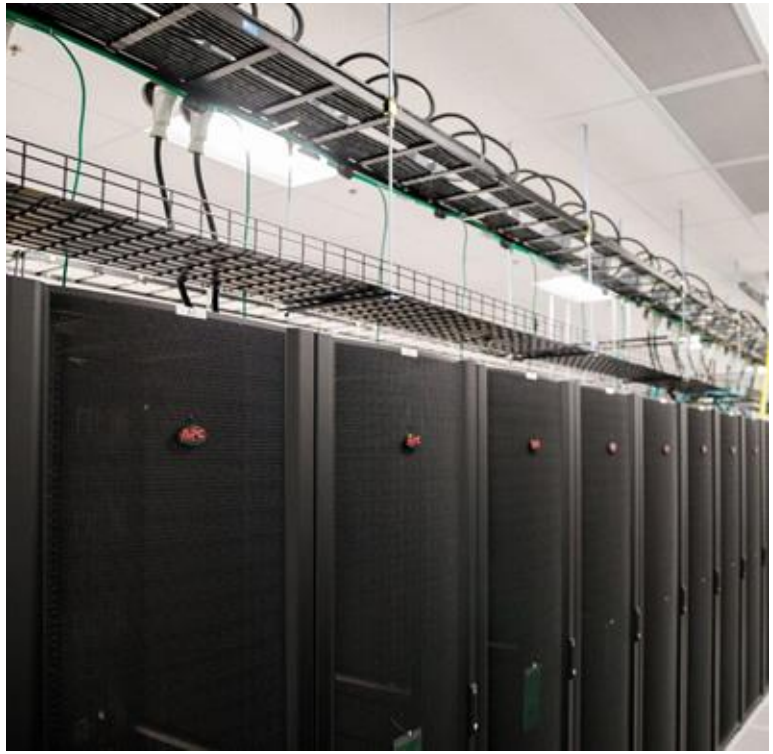
- Supercomputer
  - High-Performance Computing cluster
- Slurm
  - Job scheduler system
- Fairshare
  - Computational resources consumption
- Workflow
  - Multi-steps computational pipeline



# ASU Supercomputers

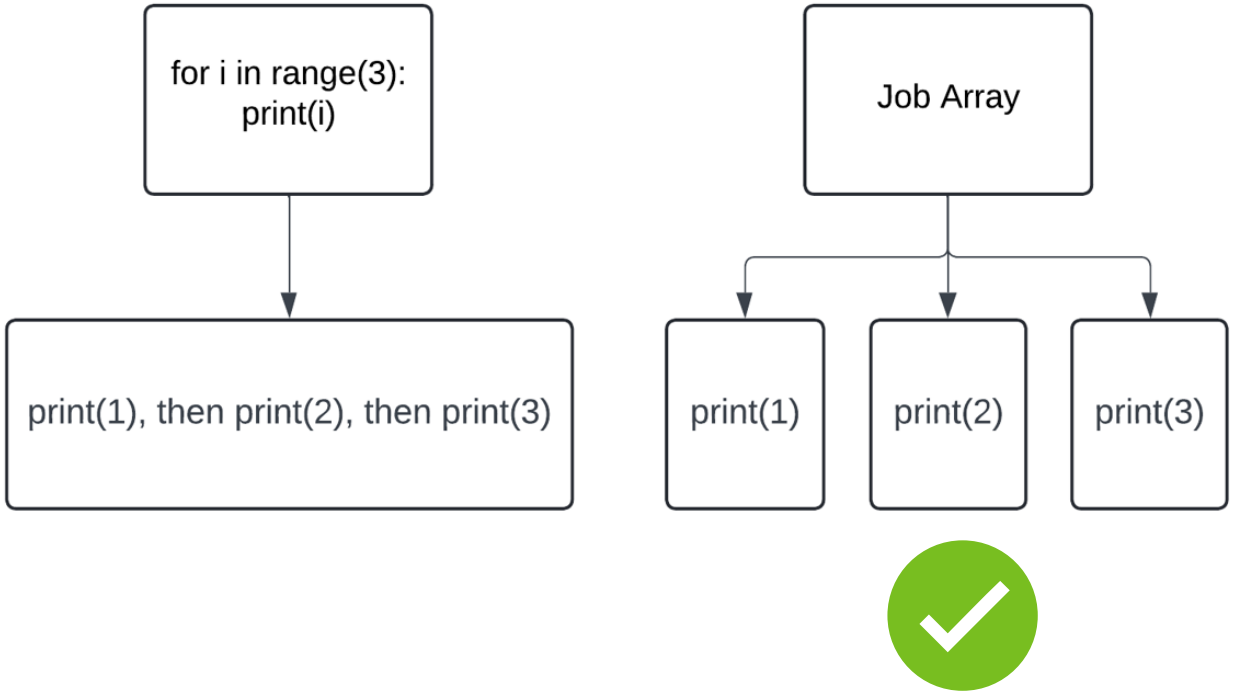
- Sol
  - Flagship, Top500
- Phoneix
  - CPU-intensive
- Aloe
  - HIPAA-compliant
- New AI cluster!

docs.rc.asu.edu



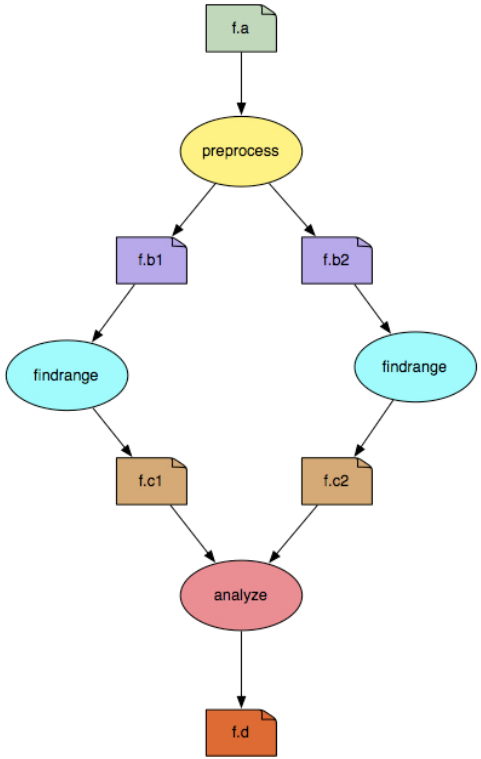
# How to Compute Efficiently?

- Main goal: Run jobs in parallel, not serial



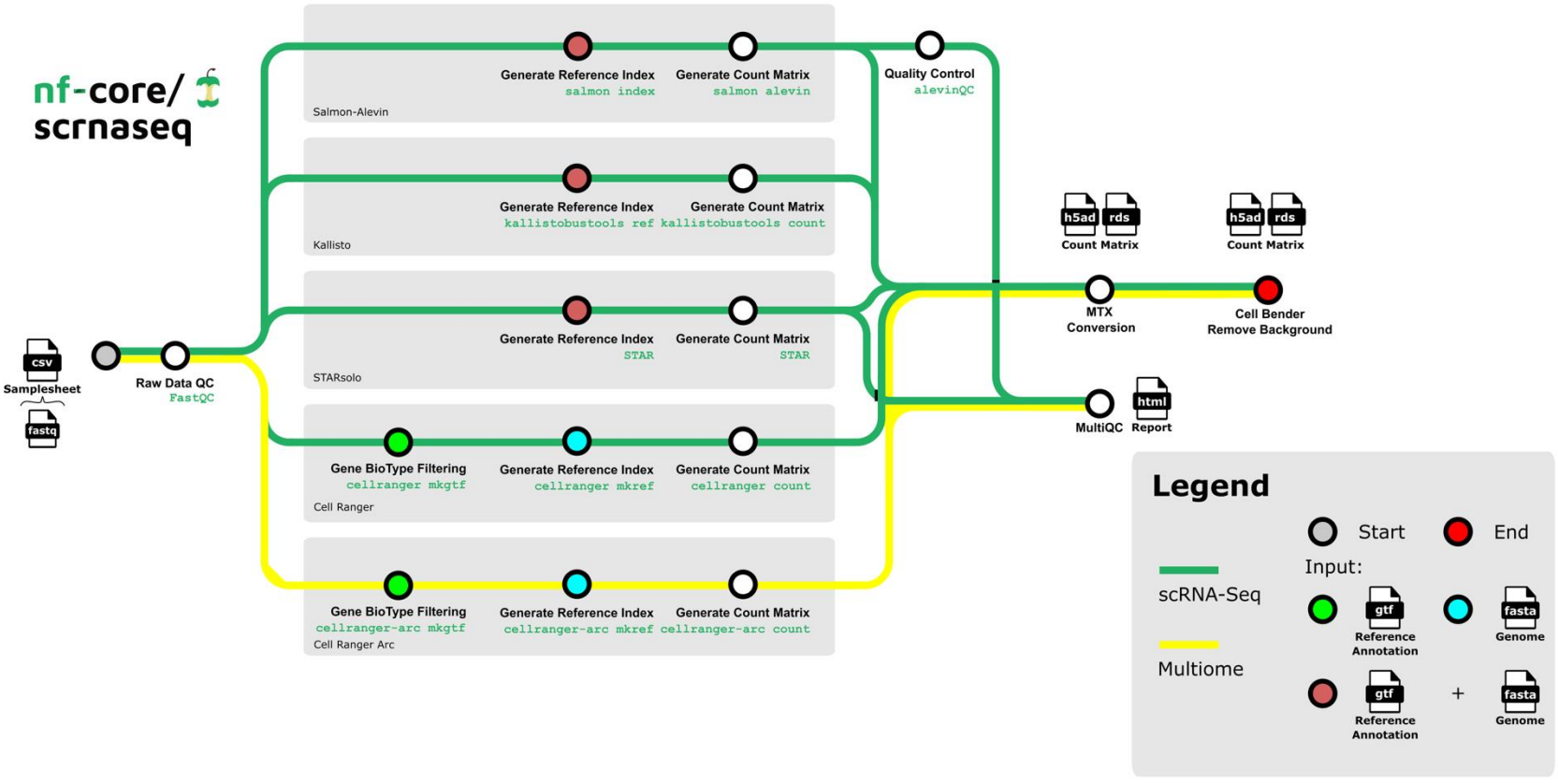
# Workflow Management System (WMS)

- Main goal: Run jobs in parallel, not serial
- DIY array vs. Slurm job array
  - Fairshare score & Pending time
- Slurm job array vs. WMS
  - Linear workflow
    - a -> b -> c -> d
  - Complicated branched workflow



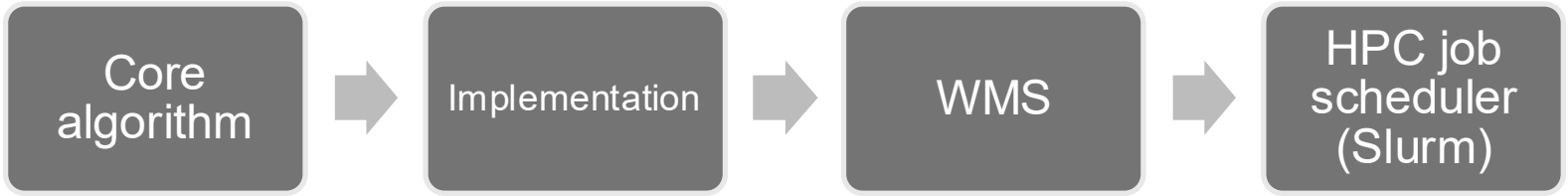

https://pegasus.isi.edu/documentation/user-guide/introduction.html#diamond-example

# Complicated Pipeline Example



# Workflow Management System (WMS)

- Main goal: Run jobs in parallel, not serial
- WMS: Communicates between your codes and the scheduler
- DSL (Domain-Specific Language): WMS syntax
- DAG (Directed Acyclic Graph): the task-dependency graph the WMS builds and walks




1. I/O
2. Resources
3. Tracking
4. Errors …

# Popular WMS

**2026 Spring**

## Snakemake

v9

Pythonic

10-1K jobs

parse-time DAG

timestamp-based

## Nextflow

v26

Groovy, Java

100K jobs

**nf-core ecosystem, lint**

**job array, cloud, container**

**run-time DAG, hashed**


## Pegasus

v5

Python, Java

1M Jobs

**multi-site**

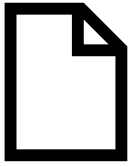
## More

***362 and growing***

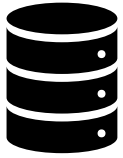
# Basic Pipeline Structure

Step description files
For actual computation

Resources configuration files
For the child jobs



Sbatch job submission script
for the main job



Input & output directories
containers, environments

Main Job

long time,
less CPU,
less memory

Child Jobs

**compute**

child job 1

child job 2

...

# Snakemake v9 Example

Snakefile: align.smk

```
rule all:
    input:
        expand("results/{sample}.bam",
sample=SAMPLES)

rule trim:
    input: "raw/{sample}.fq"
    output: temp("trimmed/{sample}.fq")
    conda: "/path/to/conda/envs/alignment"
    shell: "fastp -i {input} -o {output} -
q 20 -u 30"
```

main.sh

tmux

```
#!/bin/bash
#SBATCH -J snake_ctrl
#SBATCH -t 12:00:00
#SBATCH -N 1
#SBATCH --cpus-per-task=1
#SBATCH --mem=2G
#SBATCH -o snake_ctrl.out
#SBATCH -e snake_ctrl.err

module load snakemake

snakemake --profile profiles/slurm \
          --use-conda --conda-frontend mamba \
          --jobs 200 \
          --rerun-incomplete \
          -s workflows/align.smk
```

profiles/slurm/config.yaml

```
executor: slurm
jobs: 200
default-resources:
  time: "01:00:00"
  mem_mb: 4000
  threads: 1
cluster: "sbatch -A my_account -p general -t {resources.time} -c {threads} --mem={resources.mem_mb}"
```

# Nextflow v26 Example

align.nf

```
params.samples = "samples/*.fq"
process TRIM {
  input:
    path fq
  output:
    path "${fq.baseName}.trim.fq" into trimmed_ch
  conda "/path/to/conda/envs/alignment"
  script:
    """
    fastp -i $fq -o ${fq.baseName}.trim.fq -q 20
    """
}
```

main.sh

```
#!/bin/bash
#SBATCH -c 1
…
module load nextflow

nextflow run main.nf -c site.config -profile slurm
```

nextflow.config

```
// Nextflow config for lastz jobs
executor {
    queueSize = 10000
    retry.maxAttempt = 3
    killBatchSize = 1000000
}

process {
    executor = 'slurm'
    memory = { 4.GB * task.attempt }
    time = { 1.hour * task.attempt }
    queue = 'public'
    cpus = 1
    array = 2000
    maxRetries = 5
    errorStrategy = 'retry'
    maxErrors = '-1'
    clusterOptions = '--signal=USR2@180'
}
```

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# Challenges

# Challenges Overview

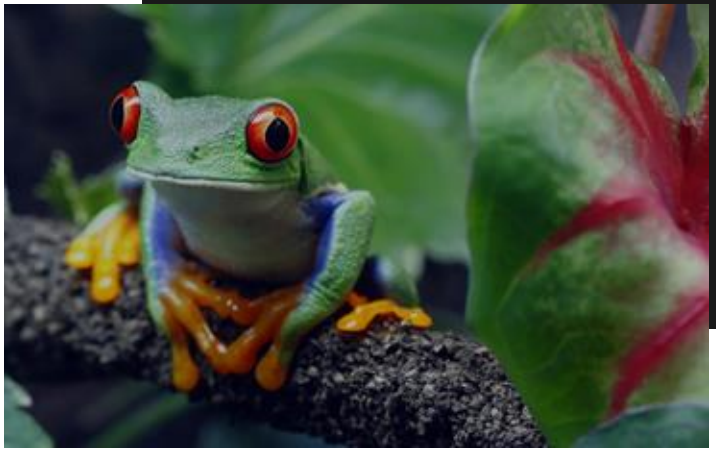
Admins

- **Scheduler interaction**
- **Fair-share impact**
- **Resource efficiency**
- **Cluster-wide utilization**

Users

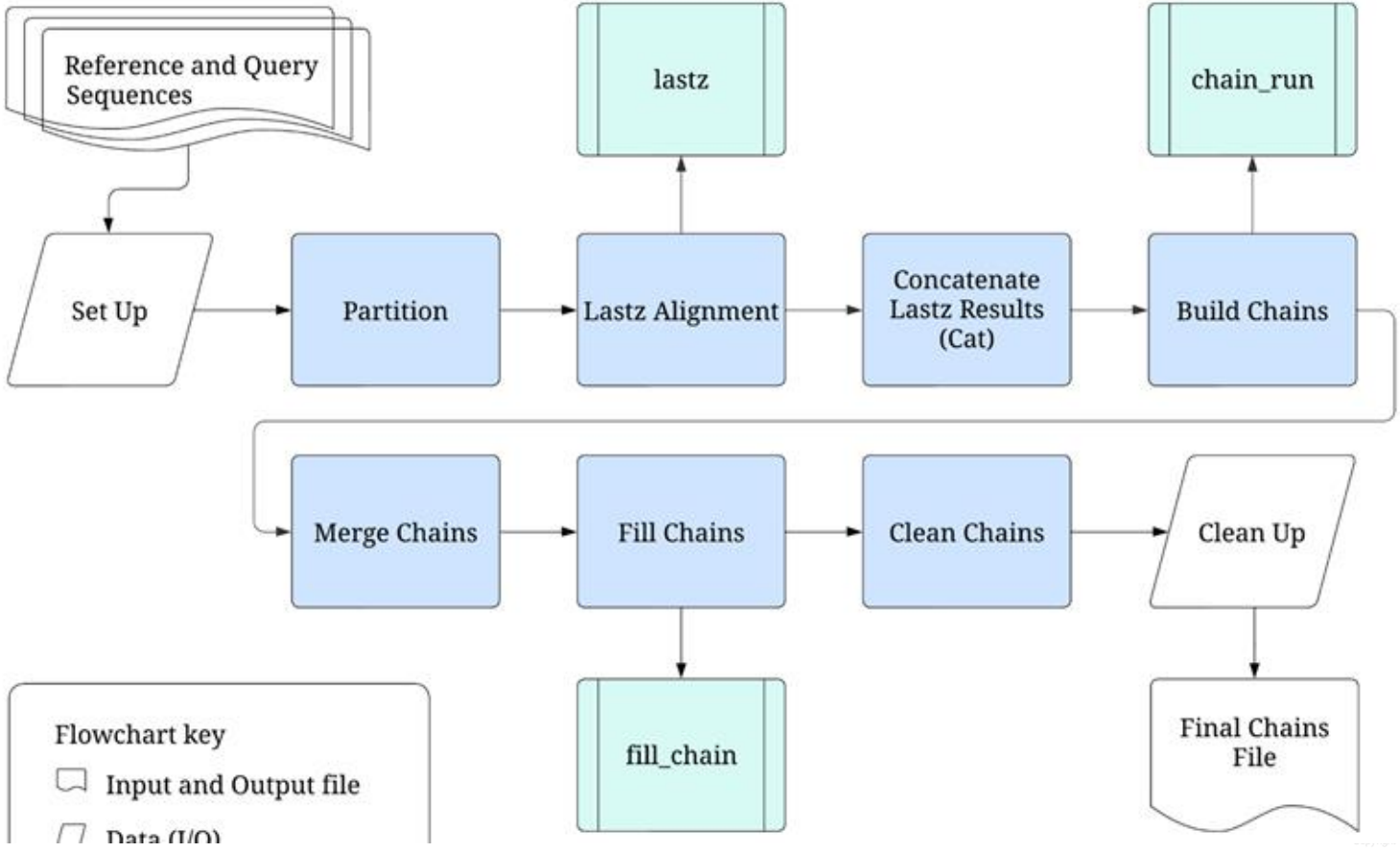
- **Over-provisioned memory & time**
- **Runaway submissions**
- **Poor locality awareness**

# Earth Biogenome Project (EBP)

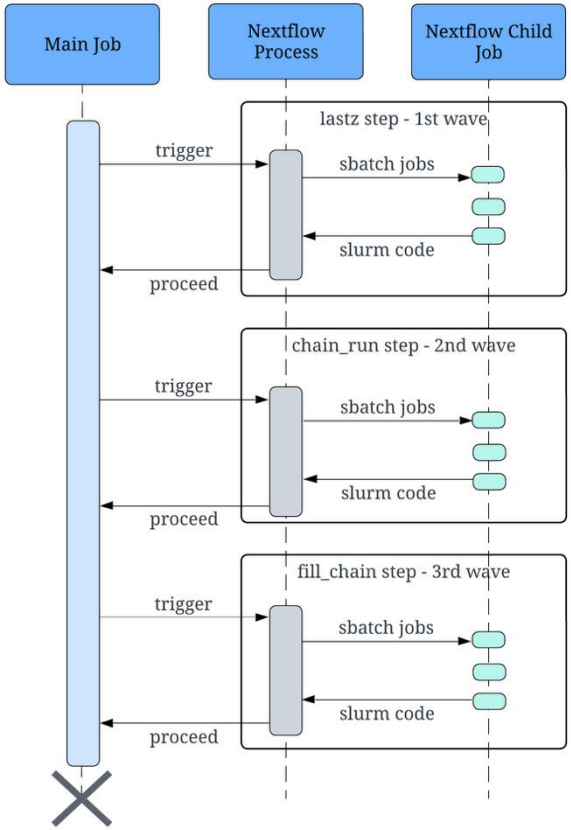


- Representative eukaryotic species
- $4.7 Billion
- 12.7% (10k)
- Compare long strings

# make_lastz_chains

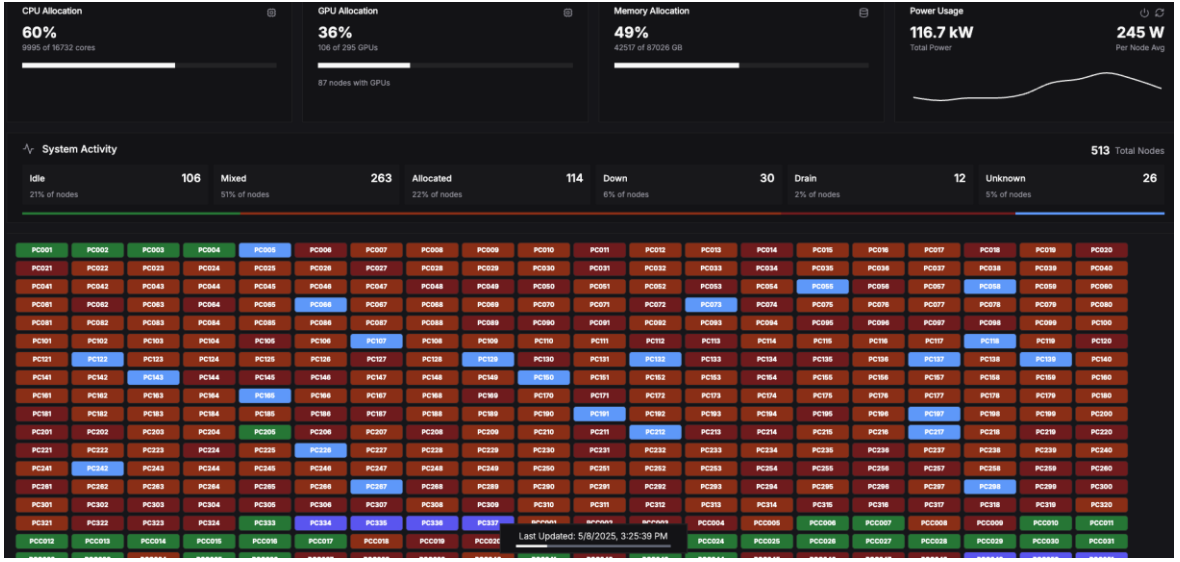


# Challenge #1: Scheduling



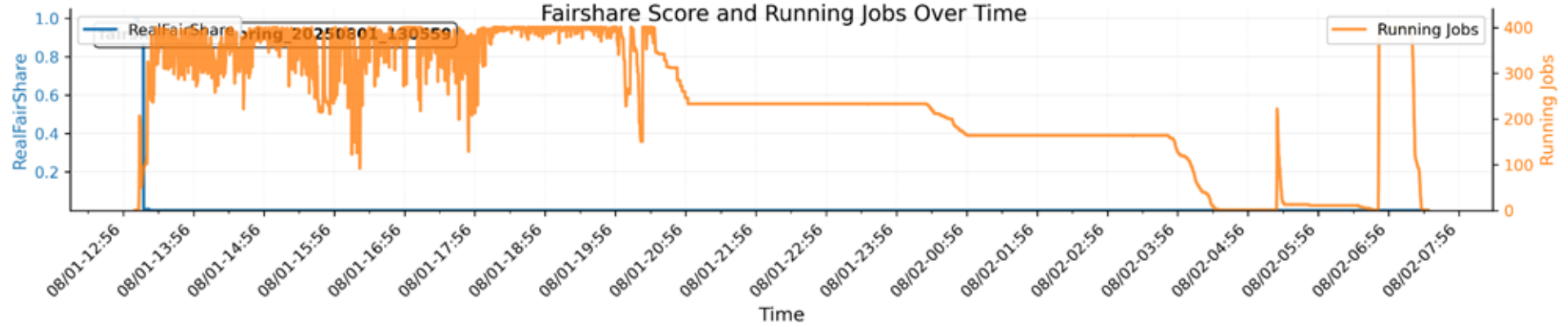

For each sample

- 3 waves per main job
- 20k ~ 200k jobs per wave

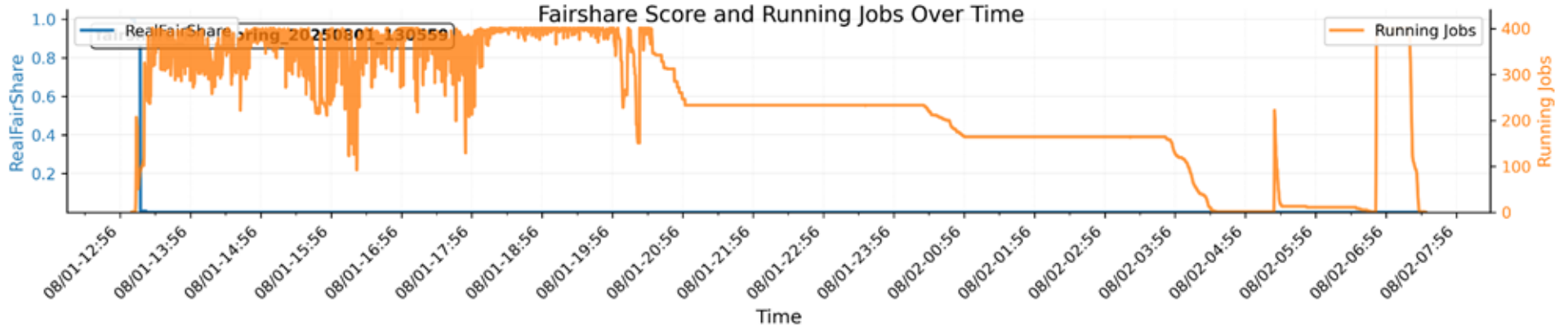


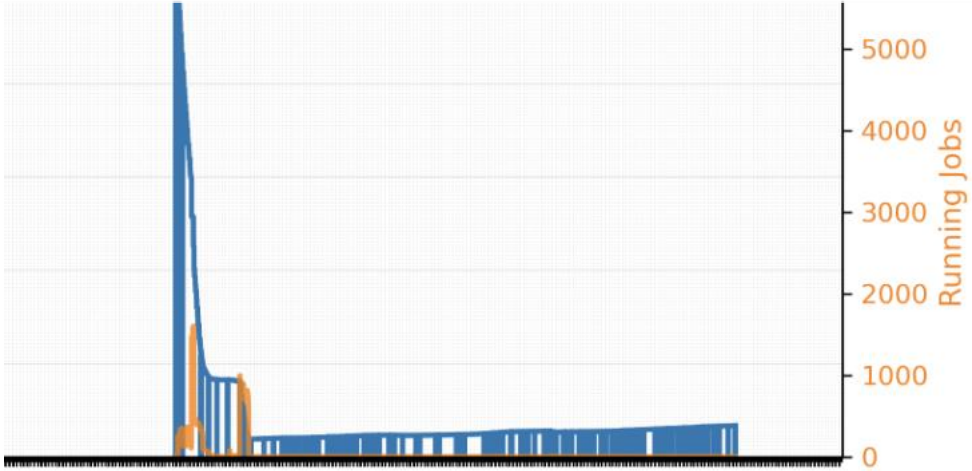
# Challenge #1: Scheduling

Fairshare score ▼ — Waiting time ▲ — Total time ▲ — Main job crashes



# Challenge #1: Scheduling



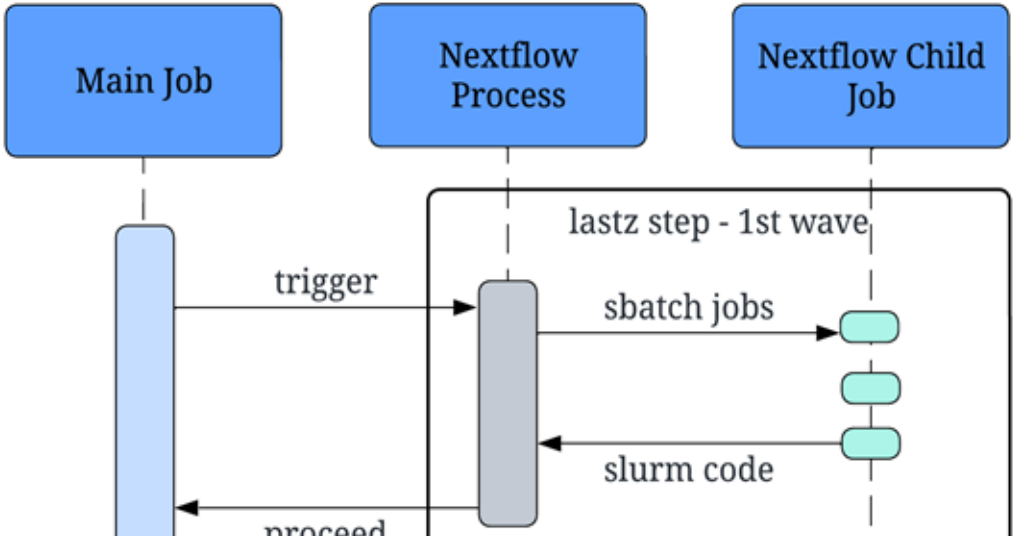




Additional Info:

Dynamic Fairshare, recovery

# Challenge #2: Failure Tolerance

- Child job: failed (OOM, timeout, etc.)
  - Slurm: sends out an error signal
    - Nextflow calls off the job step
      - Nextflow calls off the entire workflow



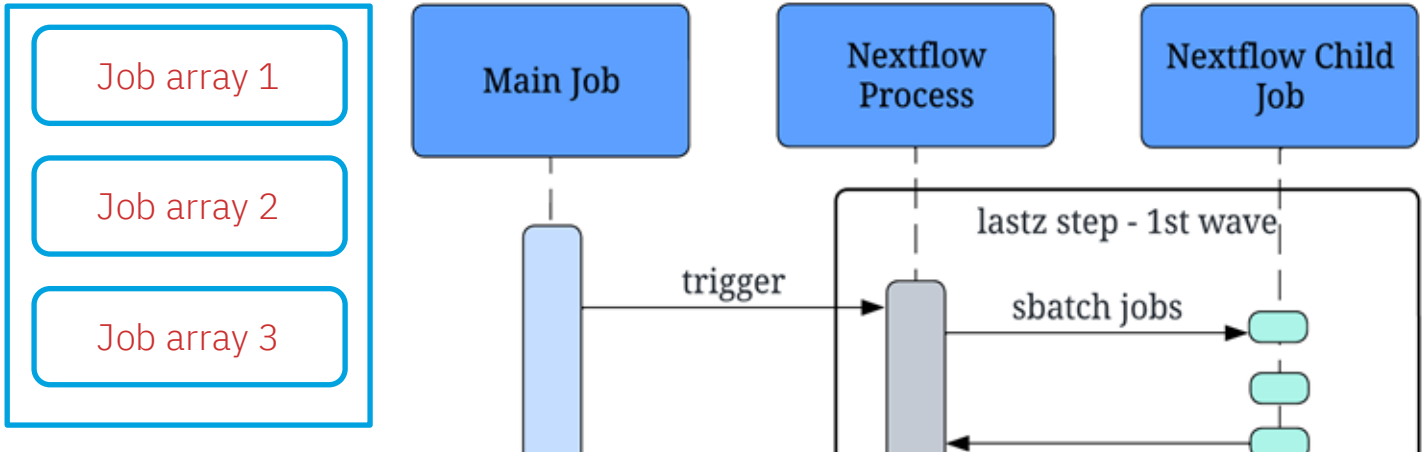
# Challenge #3: Slurm RPC limit

- Hardcoded Slurm RPC limit
- Affected by other users as well
- Runaway child jobs – no logs, not in squeue
    - Slurm log > nextflow log > child job log
- Main job stuck

# Solutions / Best Practices

- Reduce child job size (time, memory, cores) for backfilling
- Retry mechanism, error handling
- Dynamic resource allocation & selection

- Enable the Job Array feature



# Nextflow v25 Example

align.nf

```
params.samples = "samples/*.fq"
process TRIM {
  input:
    path fq
  output:
    path "${fq.baseName}.trim.fq" into trimmed_ch
  conda "/path/to/conda/envs/alignment"
  script:
    """
    fastp -i $fq -o ${fq.baseName}.trim.fq -q 20
    """
}
```

main.sh

```
#!/bin/bash
#SBATCH -c 1
…
module load nextflow

nextflow run main.nf -with-conda -with-mamba
```

nextflow.config

```
// Nextflow config for lastz jobs
executor {
    queueSize = 10000
    retry.maxAttempt = 3
    killBatchSize = 1000000
}

process {
    executor = 'slurm'
    memory = { 4.GB * task.attempt }
    time = { 1.hour * task.attempt }
    queue = 'public'
    cpus = 1
    array = 2000
    maxRetries = 5
    errorStrategy = 'retry'
    maxErrors = '-1'
    clusterOptions = '--signal=USR2@180'
}
```

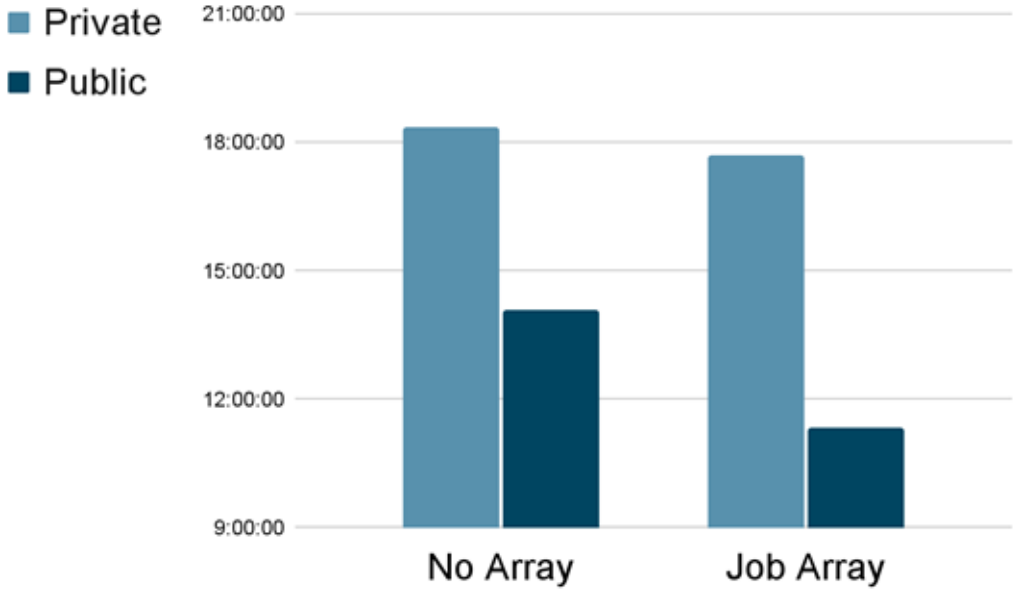
# Solutions / Best Practices

1. Reduce child job size (time, memory, cores) for backfilling
2. Retry mechanism, error handling
3. Dynamic resource allocation & selection

4. Enable the Job Array feature
5. Gracefully end the child jobs, adjust child job sizes
6. Limit child job counts, limit polling frequency
7. Out-of-band watchdog, post run assertion, slurm tuning

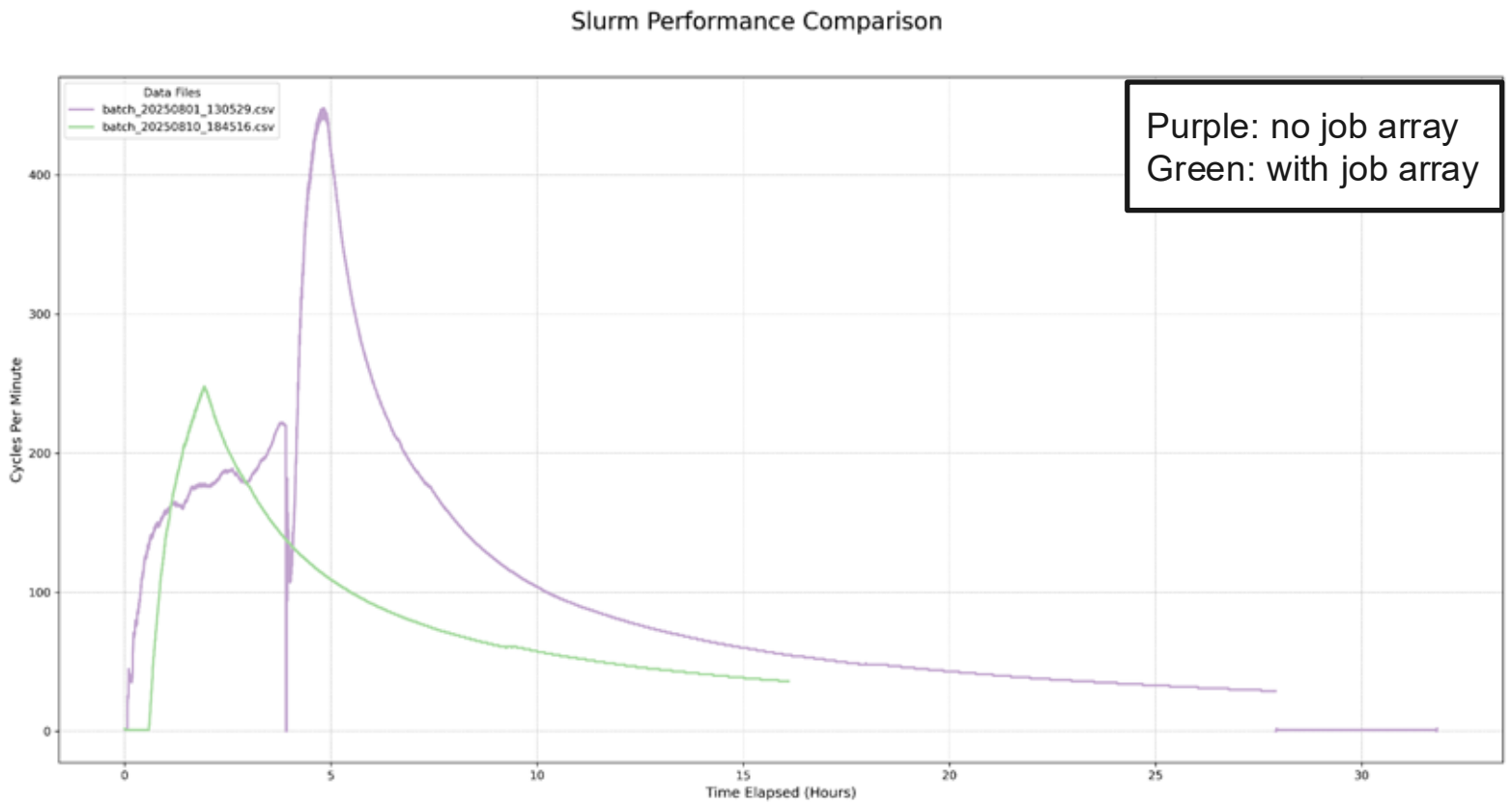
8. CPU-intensive: more cores vs. faster cores
9. Enable the Report feature: report.html, timeline.html
10. Refactor? Python-based -> Classic Nextflow -> nf-core lint

# Benchmarking – Main Job Run Time




- Workstation:
  - 4 sample/ month
- HPC:
  - 30 samples/ month
- HPC & Job Array:
  - 90 samples/ month
- Nf-core refactor
  - ~110 samples/ month

# Benchmarking – Slurm Stress



# Real Example – slurm log

```
Pipeline completed successfully!
Final chain: /data/hlewin1/tianche5/amphibian_nf/Ascaphus_truei_Lissotriton_helveticus_chains/results/final/Ascaphus_truei.Lissotri
ton_helveticus.final.chain.gz
Run time   : 1d 21h 39m 20s

executor >  slurm (954524)
[49/5ffc28] MAK…O_TWO_BIT (Ascaphus_truei) | 1 of 1, cached: 1 ✔
[5d/9e580b] MAK…_BIT_INFO (Ascaphus_truei) | 1 of 1, cached: 1 ✔
[23/0df32b] MAK…T (Lissotriton_helveticus) | 1 of 1, cached: 1 ✔
[fa/15ce17] MAK…O (Lissotriton_helveticus) | 1 of 1, cached: 1 ✔
[b7/eb5bc7] MAK… (Ascaphus_truei (target)) | 1 of 1, cached: 1 ✔
[6c/a7512a] MAK…triton_helveticus (query)) | 1 of 1, cached: 1 ✔
[a8/392e98] MAK…4_1:1198800000-1208800000) | 951750 of 951750, retries: 1358 ✔
[da/8ff20c] MAK…_1_in_250000000_260000000) | 346 of 346, retries: 10 ✔
[99/8730a9] MAK…S:CHAIN_BUILD:PSL_SORT_ACC | 1 of 1, retries: 2 ✔
[d2/56be84] MAK…INS:CHAIN_BUILD:PSL_BUNDLE | 1 of 1, retries: 3 ✔
[fa/f77c50] MAK…D:AXT_CHAIN (bundle.3.psl) | 44 of 44, retries: 4 ✔
[0a/34416e] MAK…AIN_BUILD:CHAIN_MERGE_SORT | 1 of 1 ✔
[26/534b5e] MAK…AN_CHAINS:FILL_CHAIN_SPLIT | 1 of 1 ✔
[89/b0a490] MAK…_FILLER (infill_chain_953) | 1000 of 1000 ✔
[b4/fec2a3] MAK…AN_CHAINS:FILL_CHAIN_MERGE | 1 of 1 ✔
[a1/ba1cb7] MAK…elveticus.filled.chain.gz) | 1 of 1 ✔
[23/02ad40] MAK…leaned_intermediate.chain) | 1 of 1 ✔



Completed at: 08-Apr-2026 10:41:29
Duration    : 1d 21h 47m 42s
CPU hours   : 11'304.2 (0% cached, 1.5% failed)
Succeeded   : 953'147
Cached      : 6
Failed      : 1'377
```

# Take-away

- Nextflow would be a better choice
- Main job vs. Child job
- Scheduling & Failure Tolerance
- Retry with Dynamic resource allocation
- Job array support & tuning

# Thank You!

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https://nilablueshirt.github.io/WMS-on-HPC/